Joint Genre Classification and Neural Topic Modelling using Pretrained BERT with a Variational Topic Bottleneck

Adithi P (MT2025011)

Deeksha Jain(MT2025038)

Ishita Kar(MT2025052)

Rinku Kumari Pareta (MT2025102)

Problem Definition

In this project we perform multilabel genre classification of movie or book summaries while ensuring interpretability. Given a movie or book summary the objective is to predict its genre and extract latent topics that can explain the model's prediction.

Dataset and Preprocessing

The dataset used in this work is a combination of the CMU Movie Summary dataset and the CMU Book dataset. The features include textual summaries and multi-label genre annotations. The datasets were merged by retaining only common genres, resulting in 13 genre classes. Data cleaning was performed by removing HTML tags and noisy entries. The final dataset contains approximately 49,000 samples.

Baseline Model

Baseline Approach

1. Zero-Shot Classification

A pre-trained zero-shot classification model is used to predict genres directly from text summaries. The model compares the input text with predefined genre labels and selects the most relevant one based on semantic similarity. Since it is pre-trained, it does not require training on the dataset and can generalize to unseen data. It outputs a single predicted genre for each input.

2. Topic Modeling (LDA)

Latent Dirichlet Allocation (LDA) is used to extract hidden themes from the text. The text is first converted into a Bag-of-Words representation, and LDA is applied with 32 topics. Each topic consists of a group of important words representing a theme. For example, words like *love, family, relationship* indicate a romance-related topic. Each document is considered a mixture of multiple topics.

Processing Steps

Text preprocessing: lowercase conversion, removal of numbers, short words, and extra spaces
Conversion to Bag-of-Words representation
Topic extraction using LDA
Genre prediction using zero-shot classification

Evaluation Metrics

The model is evaluated using:
Accuracy – overall correctness
Macro F1-score – balanced performance across genres
Classification Report – precision, recall, F1-score
Confusion Matrix – error analysis
Book vs Movie F1 Comparison
Class-wise F1-score visualization

Results and Analysis

The baseline model achieves moderate performance, indicating that the pre-trained model captures general semantic relationships.

Confusion Matrix:

Drama, romance, and family are often confused due to overlapping themes, while sci-fi, action, and horror are better distinguished.

Class-wise Performance:

Higher F1 scores for action, thriller, and sci-fi due to distinct vocabulary; lower scores for drama and romance due to semantic overlap.

Book vs Movie:

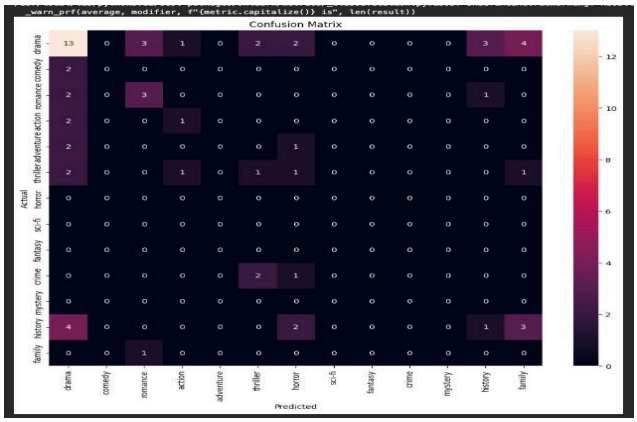
Books perform better because of richer descriptions, while movies show lower accuracy due to shorter summaries.

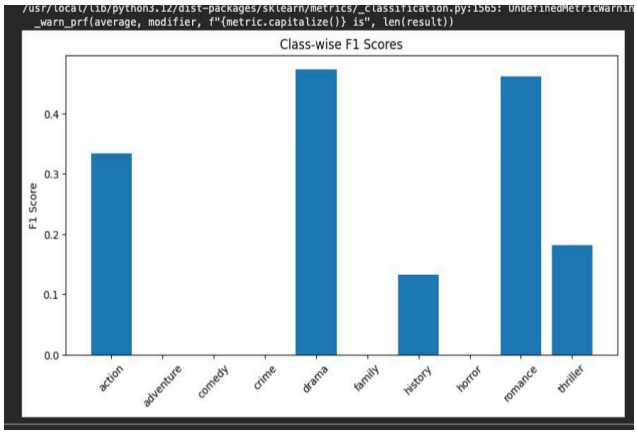
Topic Modeling:

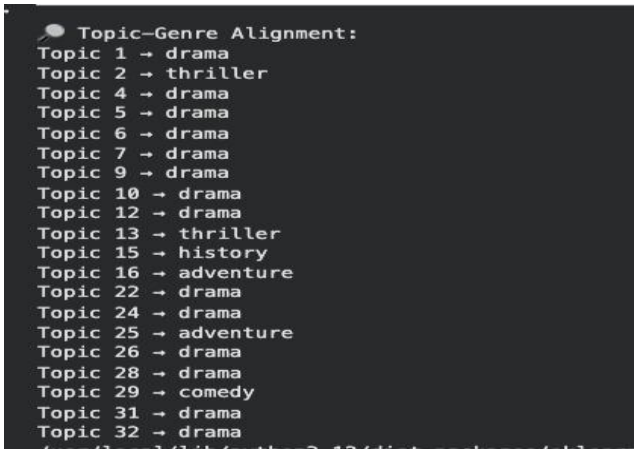
LDA extracts meaningful themes (e.g., *space → sci-fi*, *war → action/history*, *family → drama*), but topic–genre alignment is approximate and not one-to-one.

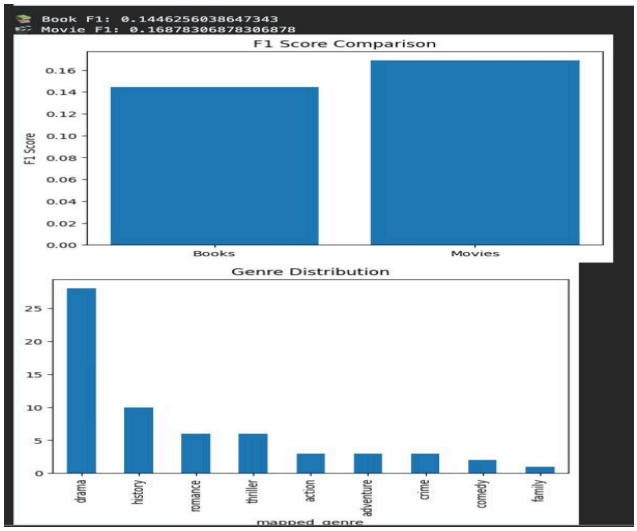
Limitations

Supports only single-label prediction
No fine-tuning of the pre-trained model
Bag-of-Words ignores context
Topic modeling and classification are independent
Limited deep semantic understanding





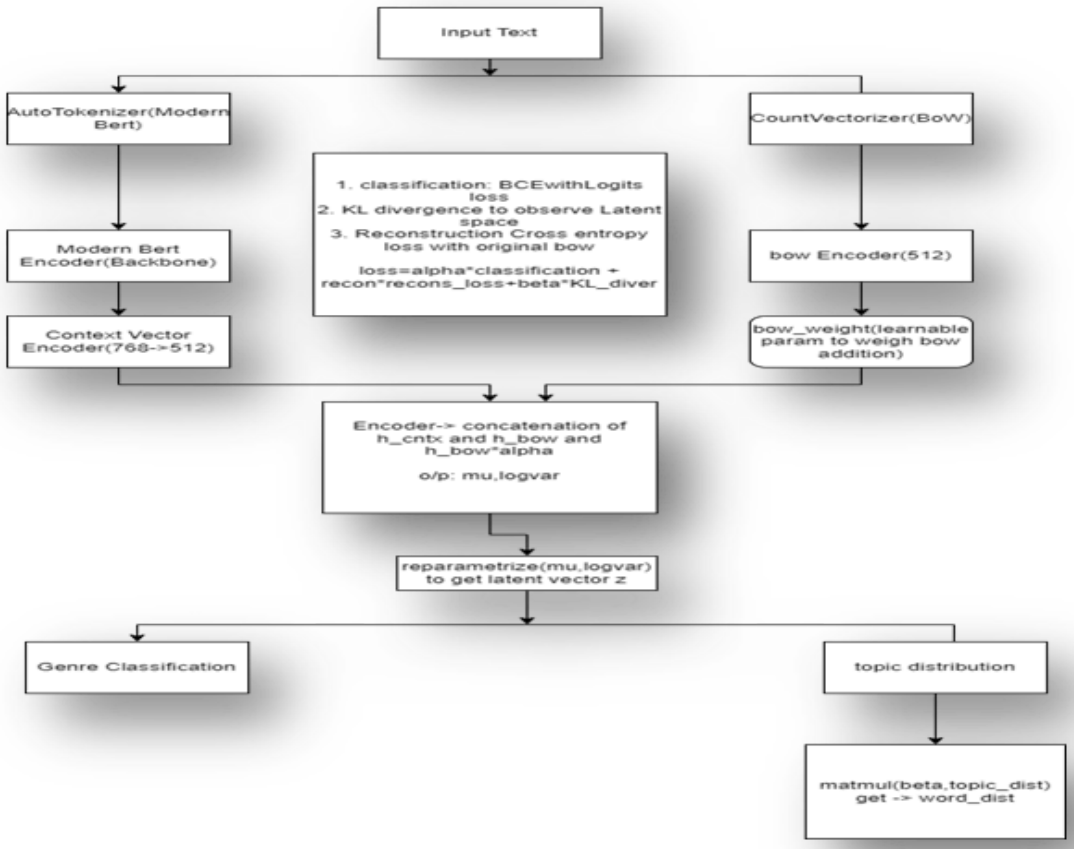




Variational Auto Encoder:

- Autoencoder is a architecture that generally represents a input as a latent vector and then uses the latent vector to reconstruct the given input. The Input is converted into a latent vector by a encoder, and the latent vector is then reconstructed back to the input image by a decoder. This is a Unsupervised learning technique.
- This principle is generally used in image generation by adding a variational component , instead of calculating a latent vector, the latent vector is assumed to be a normal distribution and the encoder component of the vae computes the mean and variance of the distribution, hence producing variation in the output image based on the input image. This technique has been the base of generative models.
- The mean and variance is used to randomly sample the latent vector from the distribution, inorder to enable differentiable gradients the sampling follows :
- **Z = mean[called mu]+variance*e,** also called the reparametrization trick.
- Here e is sampled randomly from Standard Normal distribution N(0,1) in order to separate randomness from learnable gradients.
- This technique can be used to represent the topics of the text summary of movies and books, which can help us to align topics according to the various genre present in the dataset.

Architectural Implementation using ModerBert:



1. Input text is Tokenized and we use CountVectorizer to get the word frequency in a particular text.
2. Related vocab dimension of 8000 and boW dimension is also set to 8000 for the same.
3. The Context vector of dimension NX768 and boW of dimension NX8000 are passed to context_encoder and bow_encoder to get 512 d vectors each.

4. The associated h_context and h_bow are combined to prepare contextually sensitive topics and the h_context and topic distribution so received is weighted and concatenated to predict the genre. This experiment is done as an effort to align topics and genre accordingly.
5. The topic distribution so received from latent vectors, is decoded to get the word distribution accordingly, where beta is a weighted matric that outputs the word score for given topic dsitibution.
6. This is a joint model with semi supervised learning .

Loss Function Used:

The Loss function involves the following components:

1. KL Divergence: KL divergence is used to measure how 2 different distributions are different, here we measure the latent vector distribution with Standard Normal distribution.
2. Classification loss: it is a BCEWithLogits loss used estimate each genre output as a binary cross entropy loss. This is selected since the main objective of this project is multilabel classification
3. Reconstruction loss: here is is a cross entropy loss that measures how well the reconstructed boW distribution is similar to the original boW.
4. Loss= (alpha*classification_loss)+(beta*KL_divergence)+(recon*Reconstruction loss)

Evaluation And Inference:

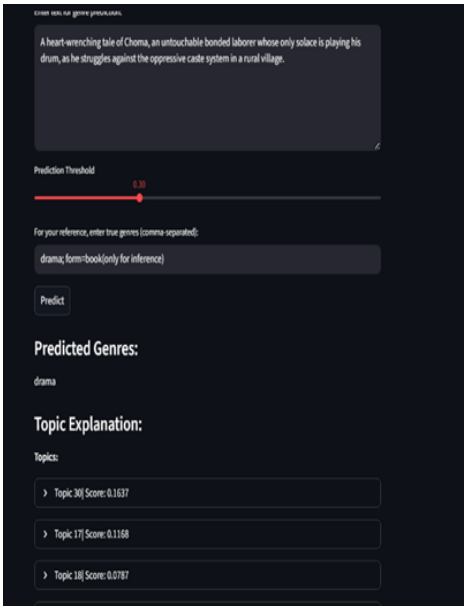
The Genre are mapped as follows:

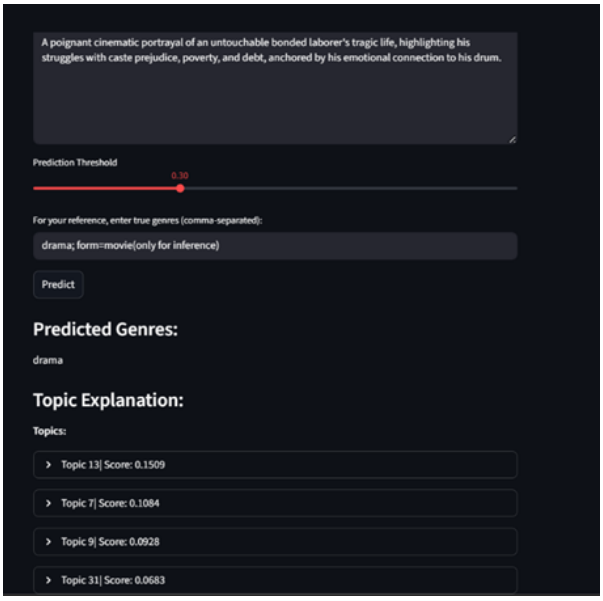
action	0
adventure	1
comedy	2
crime	3
drama	4
family	5
fantasy	6
history	7
horror	8
mystery	9
romance	10
sci-fi	11
thriller	12

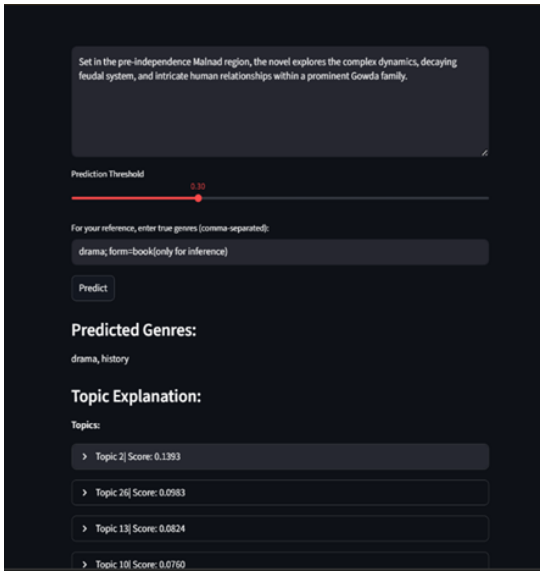
	precision	recall	f1-score	support
0	0.513015	0.684515	0.586485	691.0
1	0.338409	0.569801	0.424628	351.0
2	0.517385	0.674524	0.585596	1103.0
3	0.425505	0.635849	0.509834	530.0
4	0.752599	0.672612	0.710361	2691.0
5	0.523346	0.728997	0.609287	738.0
6	0.465409	0.654867	0.544118	452.0
7	0.333333	0.417476	0.370690	103.0
8	0.570205	0.744966	0.645975	447.0
9	0.379958	0.544910	0.447724	334.0
10	0.465812	0.599560	0.524291	909.0
11	0.636743	0.757764	0.692002	805.0
12	0.435080	0.701346	0.537020	817.0
micro avg	0.536911	0.670344	0.596253	9971.0
macro avg	0.488985	0.645168	0.552924	9971.0
weighted avg	0.561517	0.670344	0.604399	9971.0
samples avg	0.598188	0.714861	0.604502	9971.0

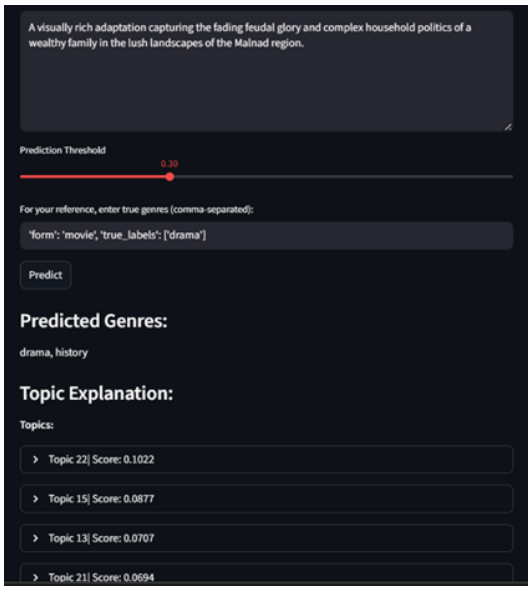
1. It can be observed that horror was a genre In our dataset with very less number of samples,
2. It can be seen our model gives a f1 score of category 7 history, where in it gives a f1-score of 0.37,
3. The Macro average of f1 score is 0.55 and weighted average is 0.60,
4. The weighted average F1-score of 0.604 confirms this joint VAE model is performing reliably across the majority of the dataset.

Inference on Movies Adapted from books:









1. We can observe that , in the first book movie pair, similar genre and different topics are predicted , this is due the style of plot description and words used.
2. In the second book movie pair, we get the drama and history as genre for drama labels and different topics could be due to usage of specific words.

<u>Inference on External Data:</u>

Genre Prediction and Topic Explanation

Enter text for genre prediction:

Hercule Poirot is vacationing in Aswan, waiting to board the steamer Karnak, which will tour along the Nile River from Shellal to Wadi Halfa. He is approached by wealthy heiress Linnet Doyle (née Ridgeway), who attempts to hire him. Linnet has recently married Simon Doyle, ex-fiancé of her former friend Jacqueline de Bellefort. Bitterly jealous, Jacqueline has taken to hounding and stalking the couple all throughout their honeymoon. Correctly suspecting that Linnet deliberately seduced Simon away from Jacqueline, Poirot refuses the commission, but privately makes an attempt to dissuade Jacqueline from "opening her heart to evil".

Prediction Threshold

0.50

For your reference, enter true genres (comma-separated):

Predict

Predicted Genres:

comedy, crime, thriller

Topic Explanation:

Topics:

Topic 18| Score: 0.2801

Top words:

confession,triggers,unaware,sooner,keith,movement,cash,lieutenant,trying,officers,woman,independent,actions,notable,unsuspecting

Present word order for the topic

cabin,mrs,mother,docks,killer,shortly,returned,pistol,seduced,victim,arrested,missing,wealthy,later,bottles,discovered,decided,married,visiting,tells,louise,jim,shots,witnessed,entering,confesses,lovers,claiming,people,honeymoon,pearl,distracted,learning,silence,dr,abu,suspected,causing,heart,night,waiting,marry,say,taken,cliff,day,does,overboard,placed,outside,ahead,socialite,present,couple,following,leg,american,collapses,brought,escape,marie,voice,unsuccessfully,executed,poirot,friend,drinks,loudly,loaded,van,caught,polish,heiress,away,went,andrew,stole,small,avoids,initially,kicks,anticipated,money,valuable,makes,raised,gun,documents,fiancé,angry,seduce,recently,attempts,goes,egypt,cousin,dead,hire,narrowly,meets,planning,reported,recovered,correctly,doyle,afraid,pushing,allow,appears,marriage,simon,faked,death,bloody,jealous,planned,injury,return,arrives,look,hysterical,refuses,necklace,spare,shoots,despite,used,attempted,bullet,killed,declares,red,treatment,lover,knows,chosen,genuine,remorse,ashore,murder,privately,realizes,missed,returns,italian,novelist,ex,claims,passengers,voyage,political,attempt,nurse,falls,romance,pair,commission,follow,crushed,second,looking,alert,young,blackmail,jacqueline,brain,leaving,elderly,fired,shot,evil,board,make,secretly,stabbed,saw,tell,daughter,came,communist,like,mr,known,room,smell,suspecting,notes,large,tried,include,trick,exposes,maid,stalking,life,miss,disappeared,physician,morning,realises,revolver,nail,bottle,plan,inheritance,colonel,did,boards,incident,wrapped,tour,reveals,immediately,approached,recovers,man,contained,opening,professional,tim,signing,string,protect,secret,fact,thought,river,confronts,deliberately,took,thief,bright,threw,race,soon,surprise

Topic 31| Score: 0.1423

Top words:

attempts,finishes,streets,career,timmy,davey,friday,mia,disgraced,sisters,dear,teachers,films,shootout,rider

Present word order for the topic

attempts,mrs,couple,daughter,allow,shot,leg,chosen,angry,life,morning,ahead,man,board,realises,y

Topic 13| Score: 0.0658

Top words:

refusing,gambling,temper,does,rocket,included,rod,quickly,enforcement,driving,plots,shipment,calling,hot,stepfather

Present word order for the topic

does,marry,following,took,incident,recovered,commission,voice,kicks,romance,river,tell,leaving,surprise,causing,confronts,visiting,raised,missed,cliff,cabin,married,planning,killed,loaded,entering,treatment,killer,wrapped,lovers,notes,inheritance,doyle,race,missing,pair,afraid,claims,recently,cousin,returned,taken,stalking,maid,known,plan,bloody,board,collapses,pearl,away,tried,genuine,money,bottle,colonel,reported,meets,came,brain,exposes,political,seduce,tour,spare,room,heart,goes,used,realises,wealthy,threw,angry,trick,honeymoon,appears,distracted,communist,silence,egypt,planned,declares,stole,life,jim,arrested,overboard,loudly,abu,smell,fired,remorse,night,murder,death,victim,follow,suspecting,shots,evil,placed,avoids,polish,present,crushed,despite,gun,jealous,arrives,mr,couple,fiancé,executed,american,andrew,italian,friend,correctly,marriage,stabbed,revolver,valuable,seduced,caught,witnessed,louise,outside,decided,second,alert,escape,attempt,make,red,small,physician,young,morning,confesses,return,nail,brought,necklace,van,voyage,fact,leg,approached,went,thought,attempts,passengers,dr,documents,mrs,waiting,did,tells,learning,secret,lover,shot,include,poirot,hire,professional,later,day,bullet,soon,jacqueline,nurse,bottles,miss,faked,knows,people,ashore,

This is an example of a crime thriller, *Death on the Nile*.
The predicted genres are **crime, thriller, and comedy**, where **crime and thriller align well with the true labels**, while comedy appears to be an incorrect or noisy prediction.

From the topic modeling analysis, the top words for each topic are presented in decreasing order of importance based on their relevance to the plot.

Topic 18 has the highest weightage, indicating strong associations with **crime and investigation**. Words such as *confession, officers, lieutenant, unsuspecting,* and *actions* reflect elements of **police involvement, hidden motives, and victim dynamics**, which are central to the narrative.

Topic 31 is the second most prominent topic, although its relevance to the plot is weak. The presence of words such as *career, teachers, films,* and *shootout* suggests a **noisy or mixed theme**, which does not strongly align with the storyline.

Topic 13 reflects a partially relevant but still weak theme. Words like *gambling, temper,* and *enforcement* hint at **character traits and crime-related elements**, but the presence of unrelated terms such as *rocket* and *shipment* indicates **poor topic coherence**.

Overall, while the dominant topic successfully captures the **core crime and investigation theme**, the other topics suffer from **low interpretability and noise**.

Limitations observed:

- Presence of **irrelevant words within topics**, indicating poor topic coherence
- **Noisy or weak secondary topics**, reducing interpretability
- **Incorrect genre prediction (comedy)** due to dataset bias or imbalance
- **Limited alignment between some topics and the actual plot**, suggesting the need for improved topic–genre connection and better training balance

Further Improvements

- Improve topic coherence by refining preprocessing (better token filtering, removing irrelevant words) and optimizing topic modeling parameters.
- Enhance topic diversity to reduce noisy or overlapping topics and ensure more distinct thematic separation.
- Strengthen topic–genre alignment so that dominant topics more accurately reflect the underlying genre of the plot.
- Address dataset imbalance by augmenting underrepresented genres to reduce biased predictions (e.g., incorrect comedy classification).
- Explore better fusion strategies between contextual embeddings, BoW features, and topic distributions instead of simple concatenation.
- Fine-tune the reconstruction–KL divergence balance to improve latent representation quality and stabilize training.
- Experiment with removing topic detachment to allow stronger interaction between topic modeling and genre prediction.

PAPER NAME- MUSIC TRANSFORMER:GENERATING MUSIC WITH LONG-TERM STRUCTURE

Ishita Kar MT2025052

1. The paper shows how Transformers can be used in generating music that exhibits long term structure. Since music has multiple dimensions like timing and pitch , along which relative differences in position matter more than the absolute position values they extend the concept of relation aware self-attention (Shaw et al. ,2018) to generate musical compositions. Another problem was that the original formulation of relative self attention required $O(L^2D)$ memory which made it impossible to train on long length sequences. To address this issue they presented a memory efficient algorithm for relative attention that reduced the memory requirement to $O(LD)$.

2. Earlier sequence models like RNN, LSTM, HMM, bidirectional LSTM have been used to model music. This paper shows the first use of Transformer to generate music having long term structure. Since in music relative positions matter more than absolute positions this work uses relative self attention mechanism(Shaw et al.). But the original formulation required a high space complexity that made it difficult to train on long sequences to capture the long range structure in longer compositions. Thats why it was necessary to address this problem and propose a memory effiecient solution.

3. In the original implementation of relative self attention in the attention score calculation a matrix S_{rel} $(L \times L)$ was introduced which was calculated as QR^T where Q is the query matrix and R is a matrix containing embedding of the relative distances between each key and query. The matrix R had a dimension of $L \times L \times D$ due to which the total space complexity became $O(L^2D)$. In this paper the authors have proposed a memory efficient solution where they don't build the R matrix. Instead they use a matrix E $(L \times D)$ which contains the relative embeddings for every possible distance. QE^T is computed. Then they apply skewing to align the logits to their correct position since each relative logit in the S_{rel} matrix $(S_{rel}[i][j])$ should be the dot product of the query in position i and the embedding of the relative distance (j-i) . Thus they have avoided the $O(L^2D)$ tensor.

4. The results are satisfactory as they have achieved significant improvement in music generation over baseline Transformer, RNN or LSTM models.

5. Even though the memory requirement was reduced to linear from quadratic , the time complexity for attention computation is still quadratic $O(L^2)$. Also they have modelled relative attention in timing and pitch. But music also has harmony,chord progression which are not fully modelled. Music has a hierarchial structure but transformer will treat everything as a flat sequence. These are some areas where the work can be improved.

Assignment 4 NLP (Topic:Contextual Word Representations: A Contextual Introduction)

Name - Deeksha Jain, Roll No. - MT2025038

a) What problem does the paper address?

- The paper actually discusses the problem of how to represent words in computers so that they capture the actual meaning of the words.
- Because in natural language, the same word can have different meanings depending on the context in which it is used.For example, the paper uses the example of "bank" which can have different meanings in different contexts. Like it can refer to the side of a river or a financial institution.The older systems gave only one fixed representation to each word and could not tell the difference between them.
- Another problem is that semantically similar words were treated as completely unrelated.
  For example, words like “peas” and “sprouts” were stored as different numbers, so knowledge learned about one word could not be used for the other, even though both are food items and related in meaning.
- Therefore, the main problem is that earlier word representations were either discrete (numbers) or static (one fixed vector per word) so they could not handle different meanings of a word in different contexts.

b) Why was it necessary to address this problem, was there any earlier work trying to address the same technology challenges ?

It was necessary to address this problem because earlier word representation methods could not capture the real meaning of words. They treated words in a very basic way, so systems could not understand similarity or context properly.The paper also gives an example that if a word like “delightful” appears in a movie review, it suggests a positive meaning, but this information cannot be transferred to similar words in older systems.There were several earlier approaches like:

- Words were stored as simple numbers, which had no meaning
- One-hot vectors were used, but they showed no relationship between words
- Distributional methods used word co-occurrence to find similarity, but they were not very accurate
- Word embeddings like Word2Vec and GloVe improved this by representing words as vectors

But all these methods had one major limitation that each word had only one fixed meaning, regardless of context. Because of this, they could not handle words with multiple meanings.

c) What unique mathematical approach does the paper use?

According to the paper, it represents words as vectors which are lists of numbers in a high-dimensional space, where similar words are closer to each other. The similarity between words is measured using cosine similarity.By using the formula $\cos(u,v) = (u \cdot v) / (||u|| \times ||v||)$ with a real worked example from a science news article shows that "bodies" scores 0.306 similarity with "objects" but only 0.134 with "astronomers," meaning bodies and objects are geometrically closer in vector space.

It also explains that word vectors are learned from large text data using machine learning, where values are adjusted using optimization methods like gradient descent.It's used to optimize a smooth, continuous objective function. Also the main idea introduced is contextual word representations. Models like ELMo use neural networks to create different vectors for the same word depending on its context. This means a word does not have a single fixed meaning, but its meaning changes based on the sentence.

d) Are the results satisfactory?

Yes,according to me the results are satisfactory because they achieve the goal of improving word understanding using context.The paper shows that models like ELMo gave strong improvements, such as about 9% error reduction in question answering and 16% in semantic role labeling. It also improved other tasks like named entity recognition and coreference resolution.Later on, the models like BERT improved performance and also reduce errors across tasks. The paper also mentions that it is rare to see such large improvements across many NLP tasks from a single idea.

e) How can the work be improved?

According to me, one major issue is bias. These models learn from large text data, so they can pick up cultural biases. For example, they may associate “doctor” with male and “nurse” with female, which is not correct. This problem needs to be handled properly.I also feel that language is not just about individual words. It also includes how words are combined in sentences. So, models should improve in understanding grammar, syntax, and overall meaning.Another improvement is in evaluation methods. The current benchmarks are not perfect and may not fully measure how well a model understands language.Finally, the models require a lot of computation, and they should be made more efficient. Also, more work is needed for low-resource languages where data is limited.

Assignment on Research Paper

NAME : RINKU KUMARI PARETA
ROLL NO.: MT2025102

TOPIC: Exploiting Cloze Questions for Few-Shot Text Classification and Natural Language Inference

(a) What problem the paper addresses

In this paper, the authors focus on a common problem in machine learning, which is the requirement of a large amount of labeled data for training models. In real-life situations, it is not always possible to collect such large datasets because labeling data takes a lot of time and effort. When we try to train models with very few examples, their performance becomes poor. So, the main problem discussed in this paper is how to make models learn effectively even when only a small amount of labeled data is available.

(b) Why was it necessary to address this problem , was there any earlier work trying to address the same technology challenges

According to my understanding, this problem is very important because most applications in NLP, like sentiment analysis or question answering, do not always have enough labeled data. Creating labeled datasets is expensive and sometimes not feasible. If we can design methods that work well with fewer examples, it can save both time and cost. It also makes machine learning more practical and usable in real-world scenarios.

(c) What unique mathematical approach the paper adopts

The paper introduces a method called Pattern-Exploiting Training (PET). The main idea is quite interesting and different from traditional methods. Instead of directly training a model for classification, the task is converted into a fill-in-the-blank type question, also known as a cloze question.

For example, a sentence like “This movie is amazing” can be converted into “This movie is [MASK]”. The model predicts a word like “great” or “bad”, and this word is then mapped to a label such as positive or negative.

The method mainly uses two important components:

1. Pattern: It converts the input text into a sentence with a blank

2.Verbalizer: It maps the predicted word to the final label

Together, these form a Pattern-Verbalizer Pair (PVP).

The process works step by step. First, the model is trained on a small labeled dataset using these patterns. Then, the trained model is used to predict labels for a large amount of unlabeled data. These predictions are not fully correct but are treated as soft labels. Finally, a new model is trained on this larger dataset.

The paper also proposes an improved version called iPET, where this process is repeated multiple times to gradually improve the model performance.

(d) Do you think the results are satisfactory , do they represent what the work was set out to be

From the experiments given in the paper, it is clear that PET performs better than traditional supervised learning methods, especially when the amount of labeled data is very small. The method was tested on different tasks such as sentiment analysis, news classification, and natural language inference.

The results show that:

The model can perform well even with very few training examples

Using unlabeled data helps improve accuracy

The iterative version (iPET) gives even better performance

Overall, the method is very effective for few-shot learning scenarios.

(e) Write your observation on how the work can be improved. Feel free to work was set out to be subsequent publication that presents a critique to this paper.

From my understanding of this paper, I found that pretrained language models already have a lot of knowledge, and the key idea is to use this knowledge properly by changing how we give input to the model. The concept of converting tasks into fill-in-the-blank questions is simple but powerful.

However, there are some limitations. The performance depends on how well the patterns are designed. If the patterns are not good, the results may not be accurate. Also, wrong predictions in the early stage can affect later training.

In conclusion, PET is a useful method for solving problems where data is limited. It also gives a good idea about how modern NLP models work and how we can improve them using simple techniques.

Name: Adithi P, Roll No: MT2025011

Paper: A DECODER-ONLY FOUNDATION MODEL FOR TIME-SERIES FORECASTING , 2310.10688

a)

1. The paper tries to build a pretrained model for time series data, similar to how LLMs like GPT work for NLP tasks. The idea is that the model should be able to do zero-shot prediction on different datasets without needing training every time.

2. Currently, most time series models like ARIMA or even neural networks are trained separately for each dataset. This paper tries to move towards a single generalized model (TimesFM) that can work across different types of time series data.

3. One challenge is that unlike NLP, time series data does not have a clear structure like vocabulary or grammar, and also large-scale datasets are not easily available. Still, the paper attempts to show that such a pretrained model is possible.

b)

1. This problem is important because there is no common model that works across many time series datasets. If such a model exists, it can reduce computation and make forecasting easier for many applications.

2. Earlier methods include statistical models like ARIMA and multivariate models like VAR, but these are not general and usually need to be tuned for each dataset.

3. Some deep learning models like PatchTST try transfer learning or semi-supervised learning, but they still do not act as a single model for all datasets.

4. There are also works where LLMs like GPT-2 or LLaMA are used for time series forecasting, but these mostly rely on fine-tuning or prompting and are not trained directly on time series data from scratch.

c)

Approach used:

a. Unlike NLP tasks using transformers, with variable length input and single pass processing with self attention, idea was to use transformers in the following work with:
    - ***Patching:*** where in they break time series data to patches similar to tokens in NLP.
    - (Enables using Tokenizer kind of a system in NLP)
    - ***Decoder Only model***: Inspired from GPT, since aim is to make generalized time series predictions, the aim will be to use previous patches and predict the next patch , similar to next token prediction task in GPT.
    - ***Longer Output patches:*** Using to predict longer patches upfron than using Auto Regressive technique unlike NLP tasks.It has been observed that long patches have shown better accuracy than Smaller token like prediction. Output patches have been longer than input.
    - ***Patch Masking***: This enables the model to see variable patch length , via random masking strategy.
    - ***Input Layers***: Use strategy of Dividing input to non overlapping patches, And a residual block which is a 1 hidden layer network where mask is combined with input and the vector of size model_dim is obtained.
    - Usage of Stacked transformer similar to GPT models are seen, with multiple attention heads,
    - Output uses decoder only models to predict patches, but input patch length need not be similar to output patch length.
    - Mean Square Error Loss function is used and calculations are done similar to Regression Problems.
    - Dataset of various types from Google Trends,Wiki Page Views, and some augmented data also used, and other source data also used.

d)

3 types of benchmark datasets were chosen accordingly:

1. **Monash:** For Monash Geometric mean based MAE for calculated for 30 different datasets of variable data length. Results observed Outperformed the Supervised ML and deep Learning models , and also compared with Zero Shot models which used GPT, and seen to outperform them as well
2. **Dart:** Univariate 8 datasets with interesting additive and multiplicative trends, The valuation is compared with baselines such as Nbeats(supervised) and also with lltime using GPT-3 like models as well. TimesFM is within statistical significance of the best models that is llmtime and seasonal ARIMA in this case.
   But no clear modelling or heirarcy of MAE thus found can be compared between the models.

3. **The Informer :** have been widely used for benchmarking various supervised long-horizon forecasting methods, focus on the other datasets in this collection (ETTm1, ETTm2. ETTh1 and ETTh2) related to electricity transformer temperatures over a two year period in 1 hour and 15 minutes granularities. TimesFM perfoms the best and the supervised PatchTST is within significance of it.
4. **Finetuning on ETT:** TimesFM finetuned thus out performed other models,

e)

1. The model currently focuses on univariate time series. In real-world problems, multiple variables are often involved, so extending it to multivariate data would improve it.

2. The model mainly focuses on zero-shot learning. While this is useful, improving fine-tuning methods could make it even better for specific tasks.